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UniGuide AI: Milestone 2

https://github.com/miranda-ruben/Digital-Information (Public)

Chosen approach: Gemini API, ‘gemini-3.8-flash’

1. Two approaches compared

AI	Gemini	Groq
Model tested	gemini-3.8-flash	openai/gpt-oss-120b
Request success rate	10/10	8/10
Average successful-request latency	24.89 seconds	7.68 seconds
Cost for 3,000 actions/month	$36	$5.85
Observed failures	Slow responses	Two HTTP 429 rate-limit errors

2. Chosen approach, three limitations, and what would make you switch

We chose the Gemini Flash API for this project, because we don't have to train the model and it is free. We asked the API already 10 questions in the Experiment above specifically about Erasmus.

The main limitations are:

Service availability: If Google's API is unavailable or the request quota is exhausted, students may not receive an answer.

Answer accuracy: The model may generate incorrect information or misunderstand university rules, so answers should be checked against official university sources.

Provider dependency: The project depends on Google's API, its pricing, its model availability and its usage limits.

What would make us switch, is an API that doesn't have such a low response quota but at the same time be free, that seems like the biggest issues for us as of right now.

4. First cost estimate

Google Gemini API pricing:

<https://ai.google.dev/gemini-api/docs/pricing>, checked 17 October 2026.

The listed paid rate for Gemini 3.8 Flash through 31 December 2026 is

$0.75 per 1M input tokens and $3.75 per 1M output tokens.

Assumption for one student action: 1,000 input tokens and 3,000 output tokens.

Calculation:

(1,000 × $0.75 / 1,000,000) + (3,000 × $3.75 / 1,000,000)

= $0.00075 + $0.01125

= $0.012 per action

At 100 actions per day for 30 days, the model-only estimate is

3,000 × $0.012 = **$36/month**

MVP Plan

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	Task	Responsible member	Target date
1	Finalize MVP scope, user journey, and acceptance criteria	Rayan Essifi	**21 Oct**
2	Select and organize official university sources for the initial use cases	Nelson Costa Alves	**25 Oct**
3	Build the basic application interface and project setup	Ruben Miranda Lopes	**27 Oct**
4	Implement the Gemini prompt and response format for administrative questions	Ruben Miranda Lopes	**30 Oct**
5	Implement the mechanism for providing relevant official information to the AI	Nelson Costa Alves	**2 Nov**
6	Connect the user interface to the AI feature and complete the main user flow	Rayan Essifi	**5 Nov**
7	Test the end-to-end flow using realistic student questions; record bugs	All three	**8 Nov**
8	Fix priority bugs, handle missing or uncertain information, and improve responses	Ruben (AI), Nelson (sources), Rayan (interface/integration)	**11 Nov**
9	Finalize setup instructions, env.example, architecture diagram, and README	Ruben Miranda Lopes	**12 Nov**
10	Verify that the application starts on another team member's laptop	Nelson Costa Alves	**13 Nov**
11	Record the three-minute demo, list known limitations, and prepare M3 summary	All three; Rayan coordinates	**15 Nov**
12	Submit the working MVP, repository link, video, and one-page summary	Team lead	**16 Nov**

Experiment

‘experiment.py’ sends ten realistic Erasmus-administration questions to the

gemini model. It records per-case success and latency, then calculates success

rate and average latency in ‘experiment_results.json’. The command used to run it is:
.\.venv\Scripts\python.exe "experiment.py"

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Results

Test case	**Response latency (seconds)**	**Result**
1. Erasmus preparation checklist	21.34	Success
2. Erasmus registration documents	18.29	Success
3. Course credit transfer	15.68	Success
4. Missing acceptance letter	38.97	Success
5. Health insurance abroad	22.50	Success
6. Accommodation questions	18.62	Success
7. Exchange budget	45.77	Success
8. Unclear university instructions	35.27	Success
9. Erasmus deadlines	18.38	Success
10. Contacting the home coordinator	14.12	Success
Average	**24.89**	**10/10 successful**

Commentary

All 10 requests returned an answer, giving a request success rate of 100%. The average response latency was approximately 24.89 seconds. The fastest response took 14.12 seconds, while the slowest took 45.77 seconds. The Gemini API successfully generated responses for all 10 test cases in this run. However, the downside is the average response time of 24.89 seconds may be slower than expected for a responsive student-support website.